

**Interaction of jaw displacement and F0 peak
in syllables produced with contrastive emphasis**

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Abstract

What is the articulatory mechanism behind the production of contrastive emphasis? How are the jaw and the larynx coordinated to produce emphatically produced syllables? It has been observed that emphasized syllables tend to involve large jaw opening and high F0. Do these two features present a conflict between the two articulators, the jaw and the larynx? If so, how do speakers resolve that conflict?

To address these questions, this study examines mandible position and fundamental frequency (F0) peaks with syllables on the low vowel /aɪ/ that vary according to whether they are contrastively emphasized or not. The stimuli were English monosyllabic words spoken in short utterances as part of read dialogues. Articulatory and acoustic data obtained by the University of Wisconsin x-ray microbeam facilities from six American English speakers were analyzed for about 90 utterances per speaker. The results show a consistent pattern of mandible position and F0 for each of the speakers as a function of the emphasis condition. For emphasized syllables, the jaw is lower and more front, while F0 is higher than for non-emphasized syllables. In addition to corroborating previous observations about larger jaw opening and higher F0 for emphasized syllables, our new finding is protrusion of the jaw in emphasized syllables.

A possible explanation of this finding is that fronting of the jaw occurs in order to achieve large jaw opening with high F0 target. The mandible position for a contrastively emphasized syllable with a low vowel is very low. To produce the high F0 for contrastive emphasis within this extremely low jaw setting may require a forward jaw, because of the biomechanical interaction between F0 control and jaw opening mediated by anatomical connections between the jaw and the larynx.

Keywords: vertical jaw displacement, horizontal jaw displacement, peak F0, syllables, contrastive emphasis, articulation

1. Introduction

What is the articulatory mechanism behind the production of contrastive emphasis? How are the jaw and the larynx coordinated to produce emphatically produced syllables? It has been observed that emphasized syllables tend to involve large jaw opening and high F0. Do these two features present a conflict between the two articulators, the jaw and the larynx? If so, how do speakers resolve that conflict?

To address these questions, this study examines jaw position and fundamental frequency (F0) peaks within syllables on the low vowel /aI/ in mono-syllabic words that vary according to whether they are contrastively emphasized or not. By studying how emphasis is implemented by our articulatory gestures, we hope to better understand the articulatory organization of speech, which must conform both to the biomechanical underpinnings of speech as well as to the principles that govern how phonological speech planning is mapped onto phonetic gestures. The conceptual framework of this investigation is according to the viewpoint that the syllable is the organizational concatenative unit of speech with jaw opening as the direct articulatory correlate of syllable prominence (e.g., Fujimura 2000).

In this framework, the extent of jaw opening for the syllable nucleus is determined by the prominence specifications to the syllable (e.g., Erickson 1998, 2002, 2003; Erickson and Kawahara 2016; Erickson et al. 1998, 2012; Fujimura, 2000, 2003; Menezes et al. 2003).¹ This prosodically-determined jaw setting affects the articulatory positions of the tongue and lips for producing specific phonological vowels through anatomical connections.

What about control of F0 associated with an emphasized syllable? F0 control and jaw opening control are in principle independent functions. Jaw opening for speech is performed by articulatory muscles, whereas F0 control is accomplished primarily by a combination of respiratory and laryngeal muscles. Generally it is thought that increased cricothyroid muscle activity results in a raised F0, and relaxing the cricothyroid results

¹ The mandible must move to achieve a segmental target gesture for the vowel in terms of a vocal tract configuration. That is, a low vowel has a low jaw position, and a high vowel, a high jaw position (e.g. Keating et al. 1994). This effect is assumed to be independent of the prosodic effects of mandible movement (Erickson and Kawahara 2016; Menezes and Erickson 2013; Williams et al. 2013).

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in a lowered F0 (e.g., Arnold 1961; Atkinson 1978; Erickson 1976, 1993, 2011; Ohala and Hirose 1969; Simada and Hirose 1971). Lowering the F0 beyond a speaker's mid F0 range may involve the extrinsic strap muscles as well (Erickson, 1976, 1993, 2011; Simada and Hirose 1971).

However, considering the anatomical complexity of the whole speech production system, there may be a certain interaction between F0 control and jaw position. Such an interaction between phonation and articulation can be understood as our control strategy for facilitating speech production, an example of which may be seen with intrinsic vowel F0 (e.g., Ladefoged 1964, Ladefoged et al. 1972, Lehiste 1970, Honda 1983, Honda and Fujimura 1990, Rossi and Autesserre 1981, Shapir 1989, Whalen and Levitt 1995, and Whalen et al. 1998). Zawadzki and Gilbert (1989) in their cineradiographic study reported a correlation between low jaw position and low F0 (with vowels with intrinsically low F0, e.g., /a/) as opposed to high jaw and high F0 (with vowels with intrinsically high F0, e.g., /i/). They suggest that jaw opening for low vowels may impose a certain mechanical influence on the vocal folds to reduce their tension, resulting in lower F0. Hirai et al. (1994) and Honda et al. (1999) proposed an extra-laryngeal physiological mechanism of F0 lowering by downward movement of the larynx along the curved cervical spine. Specifically, the MRI images in their study show that for vowel /a/, rotation of the cricoid cartilage can be facilitated by lowering of the jaw, hyoid bone and larynx, which results in lower F0. In sum, lower jaw position seems to be by default associated with lower F0, possibly because of the anatomical connection between the jaw and the larynx.

In contrastively emphasized speech, however, an opposite effect is found, with low jaw opening occurring with high F0. That is, emphasized syllables have more jaw opening than non-emphasized syllables (e.g., Erickson 1998, 2002, 2003; Erickson and Fujimura 1996; Menezes et al. 2003; Westbury and Fujimura 1989)², and also, generally, higher F0³ (e.g., Eady et al. 1986; Katz and Selkirk 2011; Lehiste 1970). High F0 in the presence of concomitant large jaw opening may present an articulatory

2 Other articulatory studies have also reported larger jaw opening for vowels that have increased prominence as the nuclear accented syllable, e.g., Beckman and Edwards 1994, de Jong 1995, Harrington et al. 2000, and Summers 1987. Other articulatory studies have also reported larger jaw opening for vowels that have increased prominence as the nuclear accented syllable, e.g., Beckman and Edwards 1994, de Jong 1995, Harrington et al. 2000, and Summers 1987.

3 The general trend in English is for contrastively emphasized syllables to have increased F0, but increased prominence can also occur with low F0 (e.g., Erickson et al. 1995, Fujimura and Erickson 2004, Menezes et al. 2002, Silverman et al. 1992). The general trend in English is for contrastively emphasized syllables to have increased F0, but increased prominence can also occur with low F0 (e.g., Erickson et al. 1995, Fujimura and Erickson 2004, Menezes et al. 2002, Silverman et al. 1992).

challenge, if large jaw opening by default results in lower F0 for anatomical reasons. Do speakers resort to a special articulatory maneuver to cope with this apparent conflict? This is one of the questions addressed in the current study.

The different patterns found in low jaw position associated both with high F0 and low F0 suggest that speakers may use different strategies of jaw movement to adjust to different prosodic requirements of the speech utterance. Generally, jaw position, viewed on a sagittal plane, reflects a combined movement of rotation and translation (e.g., Edwards and Harris 1990, Laboissière et al. 1996, Posselt 1968, Sarnat 1964, Vatikiotis-Bateson and Ostry 1995, Westbury 1988). In phonetic descriptions, vertical jaw position has served as a good indicator of jaw opening, representing a major function of the jaw gesture by the rotational component of the movement (Clark and Yallop 1995; Erickson and Kawahara 2016; Heffner 1964). On the other hand, the horizontal position of the jaw, as an indicator of the translational component of jaw movement, has been treated as a minor element of speech gesture (Clark and Yallop 1995).

One question examined in this study is the interaction between jaw position and F0, when the speaker produces contrastively emphasized syllables, given that a high F0 must be achieved within the setting of a low jaw. That is, open vowels are produced with low jaw position, which accounts for a causal mechanism of the vowel's low intrinsic F0. In emphasis, the low jaw position is re-enforced while F0 is to be raised. What are the compensatory mechanisms to realize open jaw and high F0 together? To answer this, we examine low vowels to control for various effects due to vowel height or intrinsic F0 of the vowel. Specifically, this study examines jaw setting (as defined by x-y jaw position) and peak F0 in the context of emphasized vs. non-emphasized monosyllabic words in English. The results are discussed in terms of rotational and horizontal translational components of jaw movement. Possible physiological mechanisms of jaw movement are proposed in relation to the interaction of F0 control within a specific jaw setting for a syllable, as determined by the prominence of the syllable.

2. Methods

2.1. Data Recordings

Acoustic and articulatory recordings were made using the x-ray microbeam facilities at the University of Wisconsin. For a description of the microbeam method, see Fujimura et al. (1973), Kiritani et al. (1975), Nadler et al. (1987), and in particular Westbury (1994). Spherical gold pellets (2.5-3 mm in diameter) were affixed to selected points on the tongue, lips, and jaw of the speakers (Fig. 1). Two pellets were attached to the mandible, one at the lower incisor, and another on a molar tooth (not shown in the

figure), and they were sampled at a rate of 40 samples/sec. Only the pellet attached to the mandible incisor was used for the analysis of the x-y movement of the jaw. In addition, reference pellets were affixed midsagittally to the nose bridge and to the anterior surface of the maxillary incisor. These references were used to project the articulatory pellet data onto the standard midsagittal coordinate system based on the maxillary occlusal plane (see Westbury 1991 for more detailed descriptions).

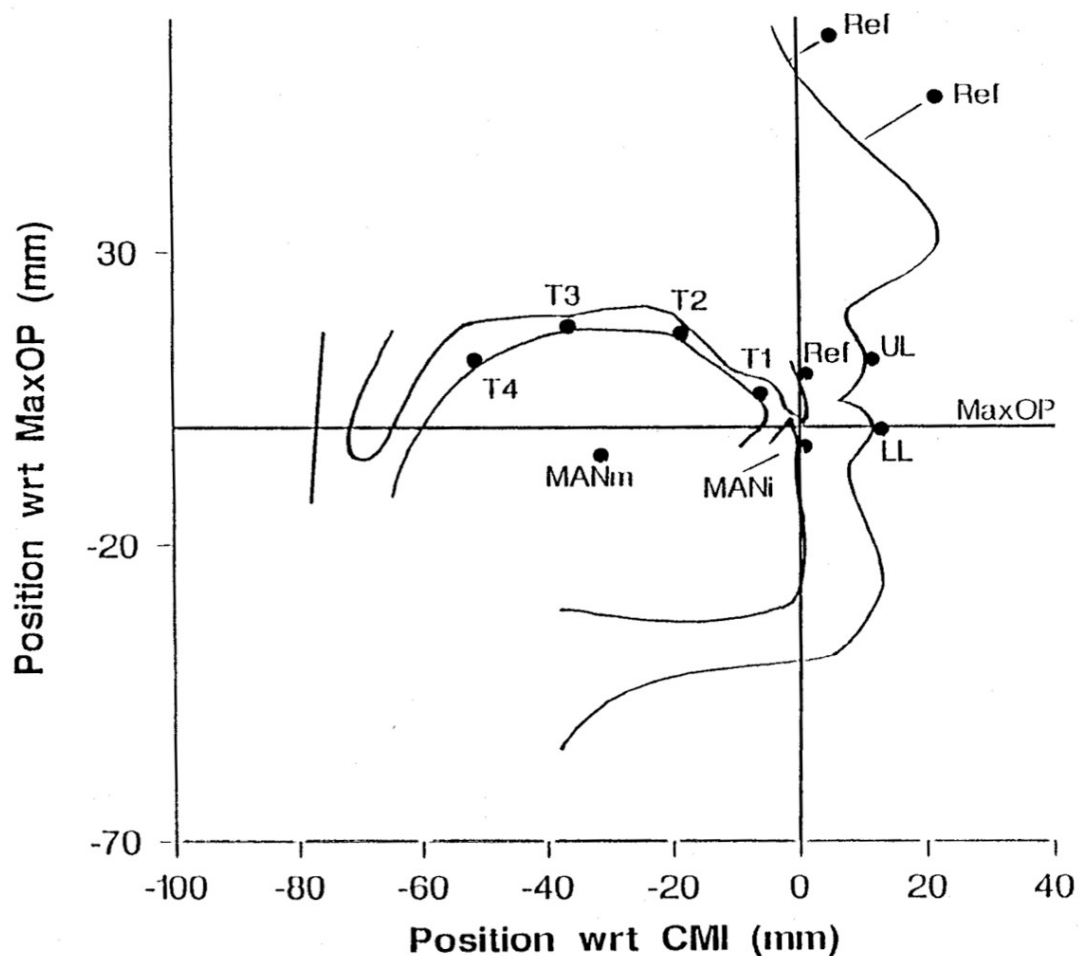


Figure 1. Placement of pellets on tongue, lips, jaw, and reference points, based on Westbury (1994: 37)

The x-axis corresponded to the intersection of the midsagittal plane and the maxillary occlusal plane, and the origin of the coordinate system was the lowermost edge of the maxillary incisor. The y-axis was normal to the maxillary occlusal plane, intersecting the plane at the origin. The coordinate value y represented the vertical

distance from the maxillary occlusal plane to the center of the pellet sphere attached to the mandible incisor, and it is always negative. In this paper, jaw opening is defined in terms of vertical mandible position; thus a maximum jaw opening corresponds to a minimum of vertical mandible position. Horizontal coordinate values represent the center of the lower incisor pellet relative to the tip of the upper incisor. Negative x-values indicate horizontal positions of the mandible pellet, which remain on the oral side of the tip of the upper incisors, and positive values indicate the amount by which the lower incisor pellet protrudes beyond the tip of the upper incisor.

2.2. Stimuli

Each of six American English speakers, (three male, three female) produced the question-answer sentence pairs like "Is it 599 Pine Street? No, it's 59FIVE Pine Street." The data-sets were collected in two stages: data for S1 (male), S2 (female), S3 (female) are from experiments by Westbury and Fujimura (1989), and for S4 (male), S5 (female) and S6 (male) are from data collected in 1996 (a subset of which was reported on in Erickson 2002—see also Menezes et al. 2003 for an experiment using a similar experimental paradigm). The same speech materials, shown in Table I, were used in these experiments.

Table I. Experiment stimuli, consisting of two digit-sequence types.

Digit Sequence Type 1:

<i>Is it 5 9 5 Pine Street? Yes, it's 5 9 5 Pine Street.</i>	(no emphasis)
<i>Is it 9 9 5 Pine Street? No, it's FIVE 9 5 Pine Street.</i>	(emphasis)
<i>Is it 5 5 5 Pine Street? No, it's 5 NINE 5 Pine Street.</i>	(emphasis)
<i>Is it 5 9 9 Pine Street? No, it's 5 9 FIVE Pine Street.</i>	(emphasis)

Digit Sequence Type 2:

<i>Is it 9 5 9 Pine Street? Yes, it's 9 5 9 Pine Street.</i>	(no emphasis)
<i>Is it 5 5 9 Pine Street? No, it's NINE 5 9 Pine Street.</i>	(emphasis)
<i>Is it 9 9 9 Pine Street? No, it's 9 FIVE 9 Pine Street.</i>	(emphasis)
<i>Is it 9 5 5 Pine Street? No, it's 9 5 NINE Pine Street.</i>	(emphasis)

The digit sequence in the answer was either "595" or "959." The utterances were randomized and read from a monitor screen, with the digit to be emphasized in capital letters, as in Table I. The emphasized digit occurred in initial, middle, or final position of the digit sequence. The utterances were also read with no emphasis when they

constituted affirmative answers such as, "Is it 595 Pine Street? Yes, it's 595 Pine Street." The digit sequences only in the answer part of the question-answer utterances were analyzed in the current study. The six speakers read each of the question-answer sequences 71 to 113 times. The number of utterances in the data sets varied due to repeated collection for records with pellet mistracking (in most cases, mistracking took place for tongue pellets, not for the mandible pellet). Mistracking resulted in the different numbers of valid utterances for mandible data across the speakers. *Ns* for each speaker are reported in Table V.

2.3. Measurements

Speech signals were recorded simultaneously with the articulatory data at a sampling rate of 22,000 samples/sec. The F0 contour was extracted using the autocorrelation-based F0 extraction program in WAVES+. A sample display of the acoustic and articulatory data for the utterance "5 9 FIVE (Pine Street)" is shown in Fig. 2. Measurements were made for each of the digits, each containing the same vowel nucleus /aI/.

As shown in Fig 2, we measured two landmark points of the syllable: the points at which the jaw was maximally open, and at which the F0 was maximally high within the voiced portion of the syllable. This measurement protocol was in keeping with the purpose of investigating the interaction of the prominence-determined jaw setting of the syllable with the peak F0 of the syllable. We did not measure mandible height and F0 at the same point in time, given the complexity of the relation between onset of muscle activity and F0 event; often, as can be seen from the figure below, maximum jaw displacement and F0 displacement do not occur simultaneously.⁴

⁴ These two articulatory events often do not occur simultaneously, but it would not be surprising if there were some synchronization between them. How they are synchronized is a topic for important research. These two articulatory events often do not occur simultaneously, but it would not be surprising if there were some synchronization between them. How they are synchronized is a topic for important research.

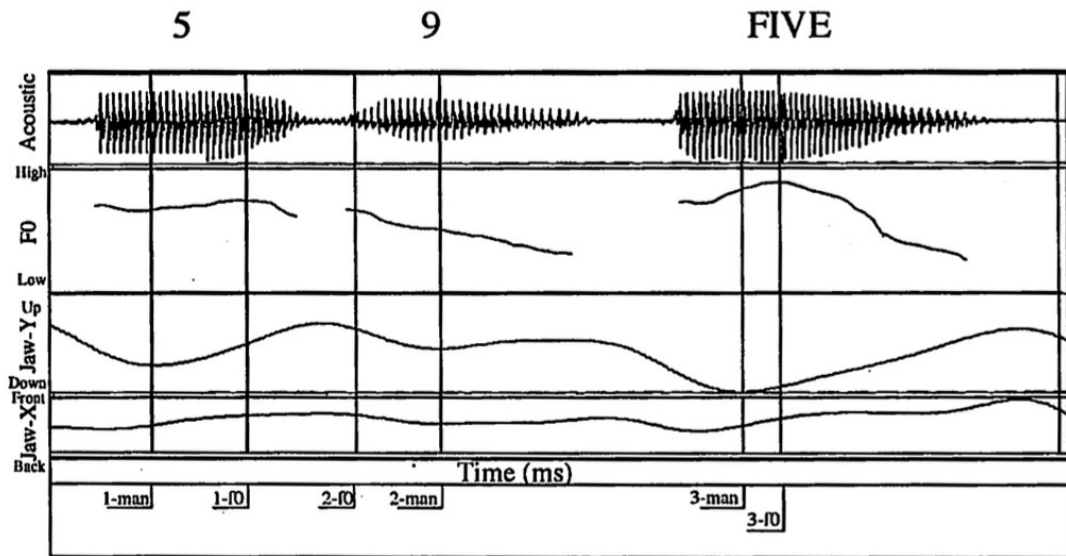


Figure 2. Display of acoustic waveform, F0 contour (50-150 Hz), jaw x-y coordinates in mm, and label window, showing where measurements were made. 1-man is the index for the minimum of mandible-y position (the point where the jaw is most open) for the first digit. The jaw-x (mandible horizontal) position was also measured at this time. The index 1-f0 is for the maximum F0 for the first digit; 2-man and 2-f0 indicate the same for the second digit; and 3-man and 3-f0 for the third digit.

According to the jaw and F0 values displayed by the WAVES+ program, measurement locations were marked manually at the lowest vertical jaw position (in reference to the maxillary occlusal plane after head movement correction), the horizontal jaw position at the time of the lowest vertical mandible position, and the peak F0 within the voiced portion of the syllable.

3. Results

3.1. Peak F0 and vertical jaw position

Table II displays maximum and minimum jaw displacement values for all emphasized and non-emphasized target syllables for each of the six speakers. From this table, one can see the variations in lowest mandible position for each speaker, as well the interspeaker variations of how much speakers open their jaws. For example, Speaker 6 opens his jaw only 2.6 mm at his minimum, whereas Speaker 2 opens her jaw 13.2 mm at minimum. Speaker 4 opens his jaw 11.8 mm at his maximum, whereas Speaker 5 can open her jaw up to 19.0 mm.

Table II. Range (maximum and minimum) for each speaker of mandible position, measured as vertical distance of the center of the lower incisor pellet in mm from the occlusal plane.

<u>Speaker</u>	<u>Maximum J-y (mm)</u>	<u>Minimum J-y (mm)</u>
S1 (m)	-16.8	-12.5
S2 (f)	-14.9	-13.2
S3 (f)	-17.6	-11.5
S4 (m)	-11.8	-5.0
S5 (f)	-19.0	-11.2
S6 (m)	-13.1	-2.6

Peak F0 was measured from the F0 contour for each of the syllables (see Fig. 2). In Fig. 3, the mean peak F0 is plotted against the mean mandible height for each of the six speakers. The x-axis indicates vertical mandible position, with the jaw most open to the right edge of the graph, and the y-axis indicates F0, with high F0 to the top. The filled circles represent the data for the emphasized syllables; and the hollow circles, the data for non-emphasized syllables.

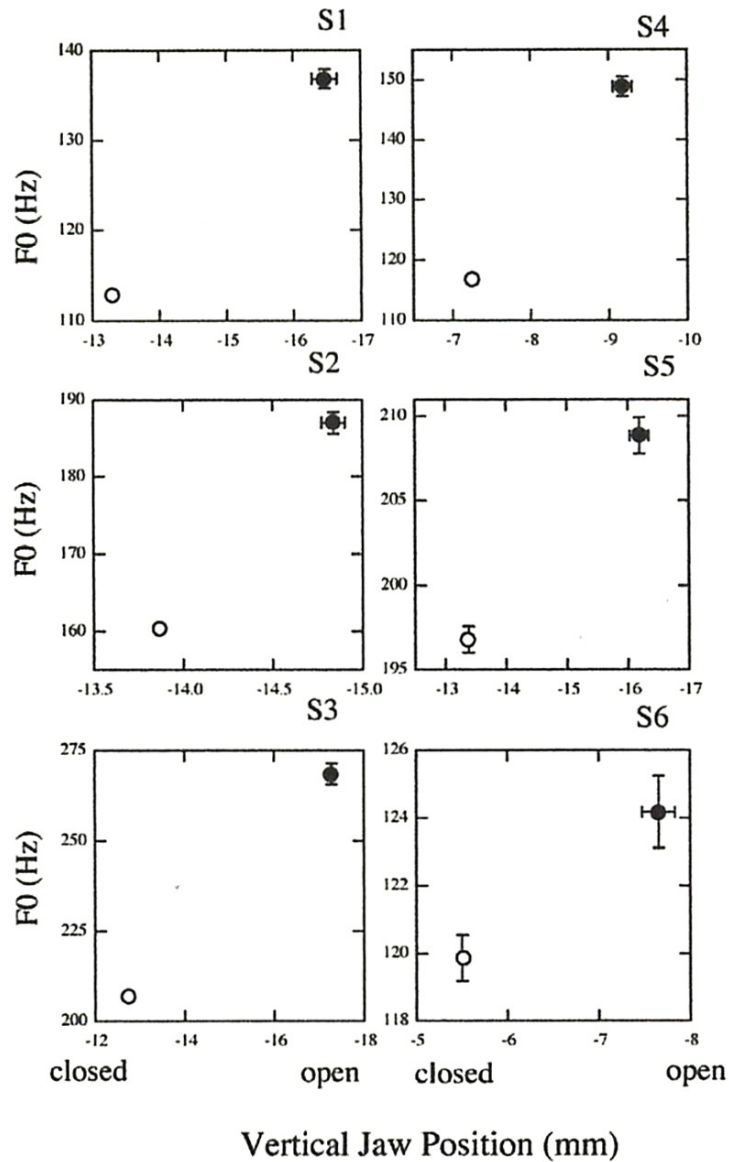


Figure 3. Vertical jaw position and F0 for six speakers. The filled circles indicate the means for the emphasized digits, and the unfilled circles, those for the non-emphasized digits. Error bars are standard errors. See Table V for the value of N.

Fig. 3 shows that emphasized syllables have both a greater jaw opening and higher peak F0 than non-emphasized syllables. A one-way ANOVA analysis indicates that both F0 and jaw opening (in the vertical direction) are significantly different between the emphasized digits and the non-emphasized digits (all at the $p < .001$ level). (See Table III, columns 2 and 4.). These results corroborate previous studies of the phonetics of contrastive emphasis: emphasized syllables involve larger jaw opening and higher F0.

Table III. F-values of the main effect of the emphasis in one way ANOVA, with vertical position, horizontal position and F0 as dependent variables. All comparisons are significant at the $p < .001$ level, except for S5, horizontal ($p = .32$)

Speaker	vertical	horizontal	F0
S1 (m)	245.63	37.06	527.65
S2 (f)	142.23	11.62	264.06
S3 (f)	949.13	164.06	453.75
S4 (m)	200.34	42.53	319.32
S5 (f)	429.39	.97	65.83
S6 (m)	155.56	51.26	11.56

3.2. Peak F0 and horizontal jaw position

The horizontal jaw position was measured at the moment of the lowest mandible position for each of these syllables. The extreme points of these horizontal positions are shown in Table IV. The "maximally protruded position" and "maximally retracted position" in the table refer to the extremely protruded and retracted values, respectively, for all the samples (including emphasized and non-emphasized) of horizontal mandible position measured at the moment of maximum jaw opening; this table thus displays the range of mandible front-back movement at the time of maximum opening for each speaker. Negative numbers indicate retraction and positive numbers indicate protrusion of the lower incisor pellet relative to the tip of the upper incisor. For five speakers, the jaw pellet on the lower incisor was found to be located at or behind the origin. One speaker, S6, showed the mandible pellet moved maximally forward by 5.7 mm from the origin, the most retracted position being 0.3 mm in front of the upper incisor, indicating Class I occlusion.⁵ Interestingly, this individual was among those exhibiting large horizontal mandibular movements.

⁵ Class I occlusion is such that "the maxillary first molar is slightly posterior to the mandibular first molar." (www.dentalcare.com) Class I occlusion is such that "the maxillary first molar is slightly posterior to the mandibular first molar." (www.dentalcare.com)

Table IV. Horizontal jaw position, at the time of maximum jaw opening during the syllable.

<u>Speaker</u>	<u>Maximally protruded position (mm)</u>	<u>Maximally retracted position (mm)</u>
S1 (m)	0.00	-6.53
S2 (f)	-5.21	-7.52
S3 (f)	-0.72	-3.52
S4 (m)	0.57	-2.04
S5 (f)	1.08	-4.28
S6 (m)	5.68	0.34

Fig. 4 plots the average peak F0 over both digit sequences against the average horizontal jaw position for each of the six speakers, emphasized and non-emphasized separately. The x-axis represents horizontal jaw position, forward being to the left, and the y-axis represents peak F0. The axes on Fig. 4 are oriented as if the speaker's head were facing the right hand side of the page, unlike Fig. 1. The origins start always at $x=0$ and $y=0$. The filled circles indicate the emphasized syllables, and the hollow circles, the non-emphasized syllables.

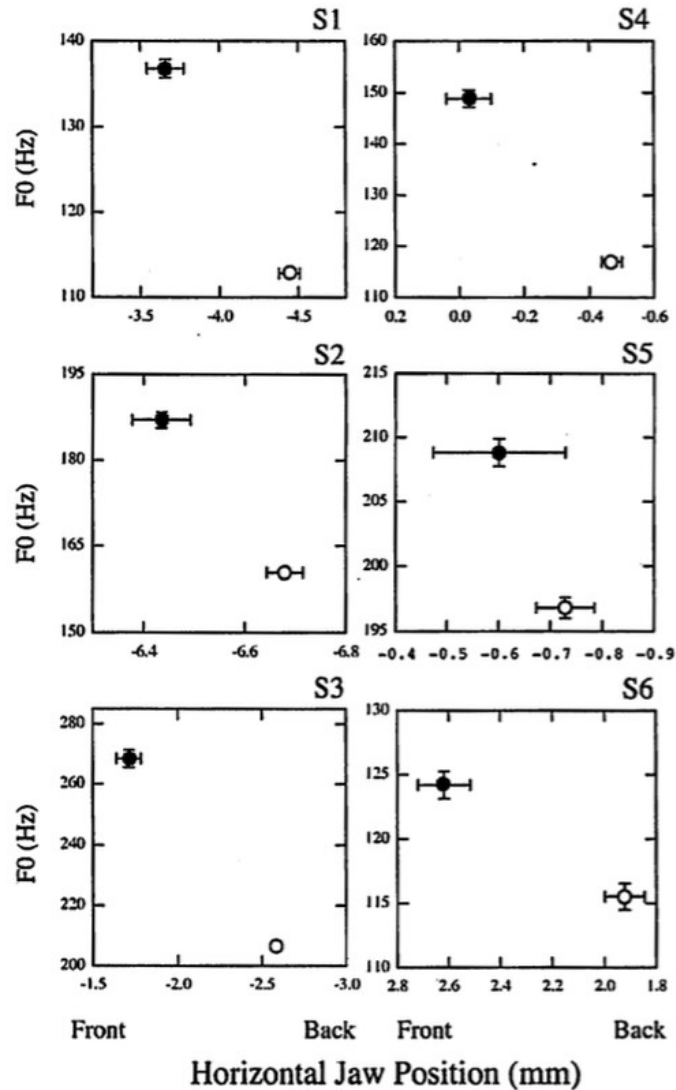


Figure 4. Mean and standard error for horizontal jaw position and F0 in six speakers. Filled circles indicate emphasized syllables, and empty circles indicate non-emphasized syllables.

Fig. 4 shows that for all speakers, emphasized syllables have both a greater jaw advancing and higher peak F0 than non-emphasized syllables. A one-way ANOVA, shown in Table III, indicates that both the F0 and horizontal jaw positions are significantly different between the emphasized syllables and the non-emphasized syllables for each of the six speakers except for Speaker 5. Speaker 5 shows a non-significant difference in horizontal jaw position between emphasized and non-emphasized digits, with large variability especially for the emphasized digits ($p = .32$). (See Table III above)

3.3. Vertical and horizontal jaw position

Fig. 5 shows the relation between the averaged horizontal and vertical jaw positions for the six speakers. The filled circles indicate the emphasized syllables, and the hollow circles, the non-emphasized syllables. For all speakers, the mean jaw position is lower and more front for the emphasized words than non-emphasized words. For Speaker 5, even though only the vertical position, not the front-back position, was statistically significant, the observed tendency was the same as the other speakers.

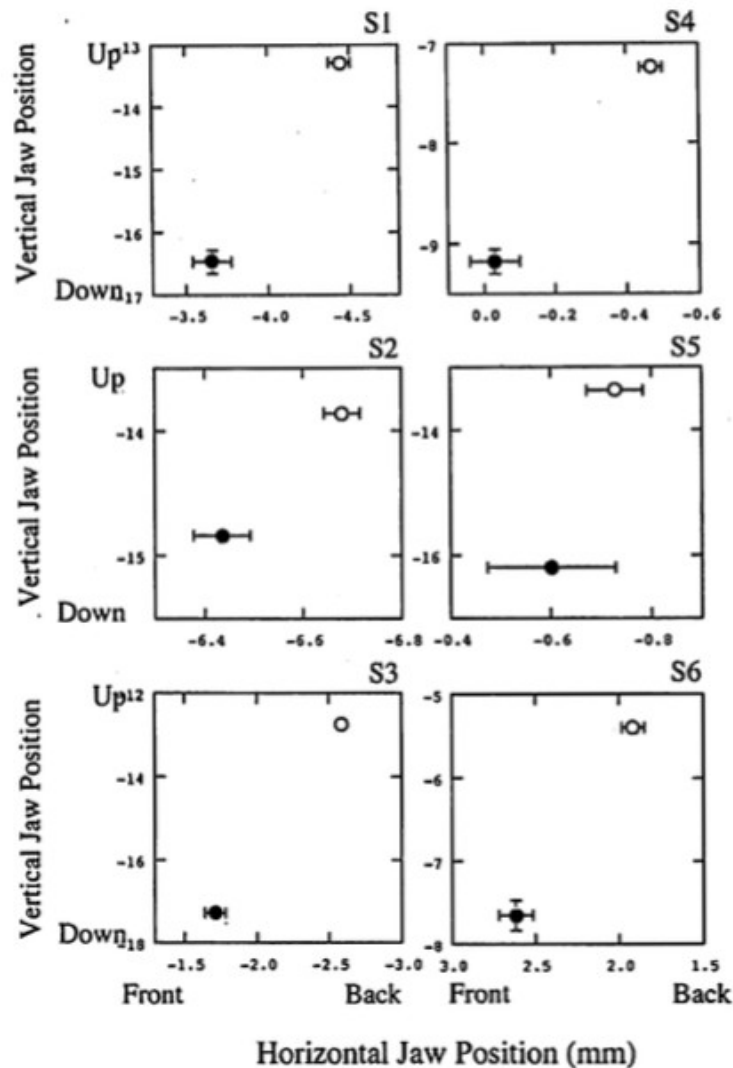


Figure 5. Mean and standard error for vertical and horizontal jaw position in six speakers.

Table V shows a summary of the mean values for peak F0, vertical jaw position, and horizontal jaw position, for each speaker for emphasized and non-emphasized syllables. The mean of vertical jaw position (relative to the occlusal plane) for each speaker ranges from -7.7 mm to -17.3 mm for emphasized syllables and -5.5 to -13.9 mm for non-emphasized syllables. The mean (over the six speakers) of mean vertical position is -13.6 mm for emphasized syllables and -11.0 mm for non-emphasized syllables. The horizontal jaw position ranges from 2.6 mm in front of the upper incisor to 6.4 mm behind the upper incisor for emphasized syllables and from 1.9 mm in front of the upper incisor to 6.7 mm behind the upper incisor for non-emphasized syllables. The mean (over six speakers) of mean horizontal positions is 1.6 mm behind the upper incisor for emphasized syllable sand 2.9 mm behind the upper incisor for non-emphasized syllables.

Table V. Peak F0 vertical and horizontal jaw positions. Standard deviations in parentheses.

Speaker	F0 (Hz)	jaw-y (mm)	jaw-x (mm)	N
S1 (m)				
Non-emph	113 (6.2)	-13.3 (1.3)	-4.4 (.8)	149
Emphasized	137 (8.0)	-16.5 (1.4)	-3.7 (.89)	58
S2 (f)				
Non-emph	160 (10.7)	-13.9 (.6)	-6.7 (.5)	174
Emphasized	187 (10.6)	-14.8 (.5)	-6.4 (.4)	56
S3 (f)				
Non-emph	206 (17.9)	-12.8 (.8)	-2.6 (.4)	170
Emphasized	269 (22.2)	-17.3 (.3)	-1.7 (.5)	58
S4 (m)				
Non-emph	117 (13.0)	-7.2 (1.0)	-4.6 (.45)	216
Emphasized	149 (14.5)	-9.2 (1.1)	-0.3 (.61)	76
S5 (f)				
Non-emph	197 (11.4)	-13.4 (.9)	-.72 (.8)	205
Emphasized	209 (9.0)	-16.2 (1.3)	-.6 (1.1)	72
S6 (m)				
Non-emph	120 (10.1)	-5.5 (1.2)	-1.9 (.7)	220
Emphasized	124 (9.8)	-7.7 (1.7)	-2.6 (.9)	87

4. Discussion

4.1. Summary of findings

Analysis of mandible position based on the x-ray microbeam data pertaining to a low vowel indicates that emphasized syllables involve (i) larger jaw opening, (ii) more forward jaw, and (iii) higher F0. The findings of more open jaw with emphasis have been reported in previous studies, as discussed in the introduction to this paper (Erickson 2002, 2003; Erickson and Fujimura 1996; Menezes et al. 2003; Westbury and Fujimura 1989). The higher F0 in emphasized syllables has also been known (e.g. Eady et al. 1986; Katz and Selkirk 2011). In addition to corroborating these observations, what is particularly new in this study is that protrusion of the jaw appears to be used along with high F0 to realize open vowels on emphasized syllables. Specifically, the results show that differential effects of rotation and translation of the jaw on F0 were shown: the jaw tended to advance for vowel emphasis with high F0. A possible mechanism for counterbalancing F0 fall by jaw translation is proposed as below.

4.2. Biomechanical account of mandible-F0 interaction in emphasized syllables

This section offers a possible hypothesis how to account for the pattern of jaw position and F0 in terms of certain anatomical constraints of the jaw and larynx that underlie the observed phenomenon. Biomechanical connections by the soft tissues exist between the jaw-hyoid and the hyoid-larynx complexes as shown in Fig. 6a.

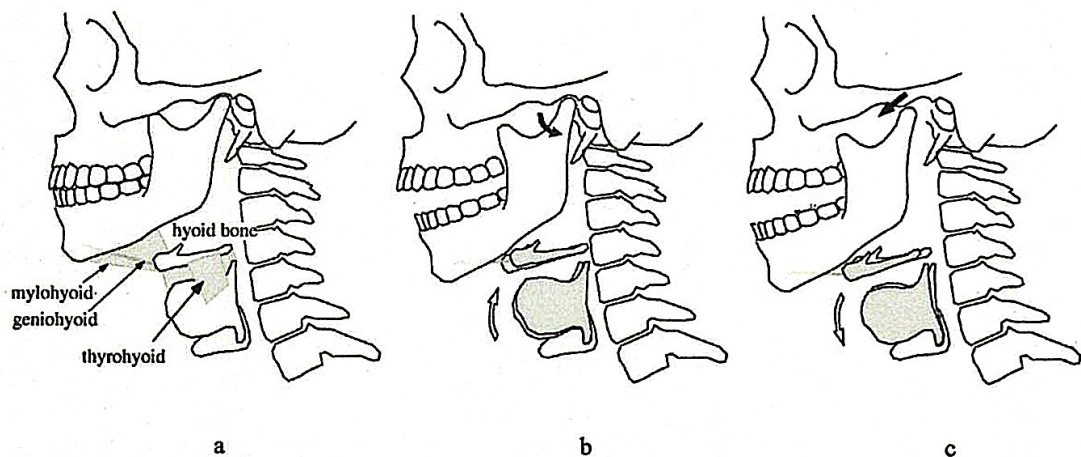


Figure 6. Possible effects of jaw rotation and translation on the larynx, (a) Anatomical relationship of the jaw, hyoid bone, and laryngeal cartilages, (b) Jaw opening by rotation can cause a reverse rotation of the cricothyroid joint, effectively shortening the vocal

folds, (c) Jaw translation pulls the hyoid bone forward to counteract the F0-lowering effect of jaw opening.

The mandible and the hyoid bone are interconnected by several muscles as well as other soft tissues: the digastric (anterior belly) (not shown in Fig 6a.), mylohyoid, and geniohyoid muscles directly, and other extrinsic lingual muscles indirectly (not shown). The thyroid cartilage is suspended from the hyoid bone by the thyrohyoid muscle and other membranes or ligaments. These soft tissue connections (shown in the figure) form the thyro-hyo-mandibular chain (Zenker 1964) and cause a mandible-larynx interaction in speech articulation. When the jaw opens, certain biomechanical effects are expected to occur with respect to the relations among these structures. The jaw opening by rotation alone (filled arrow) can cause backward translation of the hyoid bone (shaded grey) and consequent rotational separation of the thyroid cartilage from the cricothyroid around their joint (clockwise open arrow) because of the passive elasticity of the cricothyroid muscles and other tissues in the anterior portion of the larynx, as well as the horizontal friction in the more posterior area around the thyrohyoid muscle and other tissues (Fig. 6b). This change of state of the cricothyroid joint is opposite to the (vocal-fold lengthening) action of the cricothyroid muscle, and results in shortening of the vocal folds, assuming that the cricothyroid is not contracting at the same time. All other things being equal, jaw opening by rotation around the Temporal Mandibular Joint (TMJ) can lower F0 by means of the thyro-hyo-mandibular chain. This explains the tendency for the jaw lowering for low vowels to accompany low F0 as described in the literature (e.g., Hirai et al. 1994, Sawashima et al. 1982, 1983, and Zawadzki and Gilbert 1989).⁶

The findings from this study may suggest a deliberate use of jaw advancement to help increase F0 when the mandible is low. If the mandible translates forward as the jaw opens by rotation, the effect of jaw rotation on the hyoid bone is counteracted, and the consequent F0 lowering effect can be minimized or reversed. A possible mechanism for this compensatory adjustment is shown in Fig. 6c. When a wide opening of the jaw and a high F0 are both required for syllable prominence, the mandible translates forward,

⁶ Note that the jaw opening discussed here does not necessarily have to be an actively controlled mandibular abduction; if, for example, the mandible is pushed down because of the vertical thickening of the tongue body due to the posterior genioglossus contraction for front vowels as a counter action of the tongue blade elevation, the effect on the hyoid and the larynx would be the same. Note that the jaw opening discussed here does not necessarily have to be an actively controlled mandibular abduction; if, for example, the mandible is pushed down because of the vertical thickening of the tongue body due to the posterior genioglossus contraction for front vowels as a counter action of the tongue blade elevation, the effect on the hyoid and the larynx would be the same.

probably by contraction of the lateral pterygoid so that forward hyoid position for high F0 and open jaw position for enhanced magnitude of the syllable are simultaneously attained.

As the hyoid bone moves forward, the thyrohyoid muscle and membrane are stretched. Consequently, the elasticity of the tissue tends to rotate the thyroid cartilage around the cricothyroid joint in the same direction as that of the rotation brought about by cricothyroid muscle contraction. Thus, the jaw advancement, in effect, stretches the vocal folds by a chain of mechanisms, resulting in a higher F0 than otherwise, particularly when the jaw is more open for the low vowel to begin with. In other words, one way to counteract the tendency for jaw opening to accompany F0 lowering would be to protrude the jaw. The average distance between the instances of the most protruded and most retracted positions as measured at the time of maximum jaw opening, observed for the six speakers in this study, was 3.6 mm, ranging from 6.5 mm (S1) to 1.5 mm (S4) (see Table IV).⁷

5. Summary

This paper examined how peak F0 in a syllable interacts with the prosodically-determined jaw setting of emphasized low vowels. The results showed that the mandible moves forward, in addition to being lowered. This forwarding gesture may be required to achieve both large jaw opening and high F0. The biomechanics of the two biomechanical systems, the jaw and the larynx, under the anatomical constraints of their mutual interaction, accounts for the horizontal jaw movement when the jaw setting is extremely low, as required for a contrastively emphasized syllable on a low vowel. The findings from this study about the biomechanical interaction between the articulatory and phonatory organs can also be used toward developing a realistic biological model of speech production (e.g., Hirai et al. 1995), which is a necessary ingredient of a comprehensive model of phonetic organization (Fujimura, 1992, 1994, 2003).

⁷ In dental studies, the maximum range of human jaw translation is about 10 mm (Posselt 1968). This value is obtained by a measurement with a closed jaw from the maximally retracted to maximally protruded position; in our study, range of horizontal jaw position measurements are made with an open jaw. Although the measurements in our study and the dental measurements are not equivalent, the ranges of displacement of the jaw observed for our speakers seem to occupy a fair portion of the physiologically feasible range of jaw translation. In dental studies, the maximum range of human jaw translation is about 10 mm (Posselt 1968). This value is obtained by a measurement with a closed jaw from the maximally retracted to maximally protruded position; in our study, range of horizontal jaw position measurements are made with an open jaw. Although the measurements in our study and the dental measurements are not equivalent, the ranges of displacement of the jaw observed for our speakers seem to occupy a fair portion of the physiologically feasible range of jaw translation.

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Editor
Speech and Communication

Dear Sir,

We are submitting our manuscript entitled, "Interaction of jaw displacement and F0 peak in syllables produced with contrastive emphasis" by Donna Erickson, Kiyoshi Honda, Osamu Fujimura and Shigeto Kawahara.

Please let us know if there are any questions. We look forward to hearing from you,

Sincerely,

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